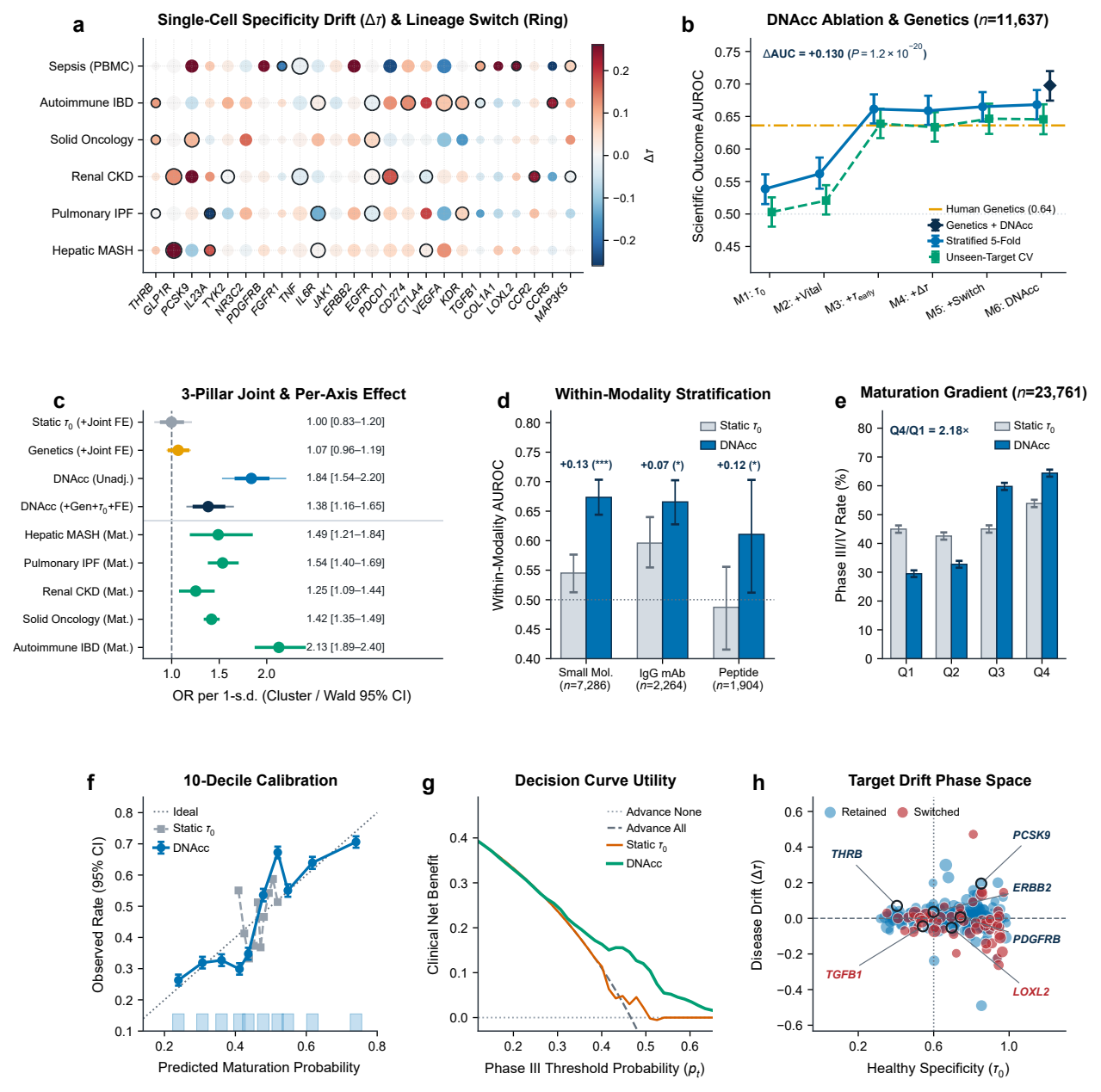

a
Single-Cell Specificity Drift (Δτ) & Lineage Switch (Ring)
Sepsis (PBMC)
Autoimmune IBD
Solid Oncology
Renal CKD
Pulmonary IPF
Hepatic MASH
THRB GLP1R PCSK9 IL23A TYK2 NR3C2 PDGFRB FGFR1 TNF IL6R JAK1 ERBB2 EGFR PDCD1 CD274 CTLA4 VEGFA KDR TGFB1 COL1A1 LOXL2 CCR2 CCR5 MAP3K5
Δτ
0.2
0.1
0.0
−0.1
−0.2
b
DNAcc Ablation & Genetics (n=11,637)
ΔAUC = +0.130 (P = 1.2 × 10−20)
Scientific Outcome AUROC
0.75
0.70
0.65
0.60
0.55
0.50
0.45
Human Genetics (0.64)
Genetics + DNAcc
Stratified 5-Fold
Unseen-Target CV
M1: τ0
M2: +Vital
M3: +τearly
M4: +Δτ
M5: +Switch
M6: DNAcc
c
3-Pillar Joint & Per-Axis Effect
Static τ0 (+Joint FE)
Genetics (+Joint FE)
DNAcc (Unadj.)
DNAcc (+Gen+τ0+FE)
Hepatic MASH (Mat.)
Pulmonary IPF (Mat.)
Renal CKD (Mat.)
Solid Oncology (Mat.)
Autoimmune IBD (Mat.)
1.00 [0.83–1.20]
1.07 [0.96–1.19]
1.84 [1.54–2.20]
1.38 [1.16–1.65]
1.49 [1.21–1.84]
1.54 [1.40–1.69]
1.25 [1.09–1.44]
1.42 [1.35–1.49]
2.13 [1.89–2.40]
1.0
1.5
2.0
OR per 1-s.d. (Cluster / Wald 95% CI)
d
Within-Modality Stratification
Within-Modality AUROC
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
0.40
Static τ0
DNAcc
+0.13 (***)
+0.07 (*)
+0.12 (*)
Small Mol. (n=7,286)
IgG mAb (n=2,264)
Peptide (n=1,904)
e
Maturation Gradient (n=23,761)
Q4/Q1 = 2.18×
Static τ0
DNAcc
Phase III/IV Rate (%)
80
70
60
50
40
30
20
10
0
Q1
Q2
Q3
Q4
f
10-Decile Calibration
Ideal
Static τ0
DNAcc
Observed Rate (95% CI)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.2
0.4
0.6
0.8
Predicted Maturation Probability
g
Decision Curve Utility
Advance None
Advance All
Static τ0
DNAcc
Clinical Net Benefit
0.4
0.3
0.2
0.1
0.0
0.2
0.3
0.4
0.5
0.6
Phase III Threshold Probability (pt)
h
Target Drift Phase Space
Retained
Switched
THRB
PCSK9
ERBB2
PDGFRB
TGFB1
LOXL2
Disease Drift (Δτ)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.2
0.6
1.0
Healthy Specificity (τ0)